

Re-processing Perturb-Seq datasets from FASTQ

First Results for the Unified Pipeline

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All Hands Meeting

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Why re-process from FASTQ?

- ▶ **Consistency:** scPerturb datasets are usually deposited as processed H5AD files, but they come from different pipelines, references, and filtering choices.
- ▶ **Comparable outputs:** Re-processing gives us a common count matrix layout, common metadata expectations, and a consistent way to attach guide counts.
- ▶ **Quality control:** Raw reads let us apply the same checks across studies instead of inheriting every author-specific decision.
- ▶ **Catalogue expectation:** Users expect harmonised data in the Perturbation Catalogue, not only links to heterogeneous processed files.

Why Kallisto + Bustools?

- ▶ **Speed:** Kallisto + Bustools is much faster than Cell Ranger while giving very comparable expression quantification.
- ▶ **Resource fit:** We have adequate, but not unlimited, compute and storage. The pipeline must work within those limits.
- ▶ **Full catalogue scale:** We want to process all datasets we have, and be able to reprocess them if the pipeline improves.
- ▶ **Future-proofing:** New perturb-seq datasets are expected to be very large, so the pipeline needs to scale well rather than depend on throwing more hardware at each run.

Pipeline Status

- ▶ The unified FASTQ-to-H5AD pipeline is now implemented in Nextflow and runs on the EBI Slurm cluster with Singularity.
- ▶ The first full dataset has been processed end-to-end: Nadig 2025 Jurkat.
- ▶ The output is a single compressed `experiment_final.h5ad`.
- ▶ Gene expression counts are stored in `.X`; guide counts are stored in `.obsm['guides']` with guide names in `.uns['guide_names']`.
- ▶ Per-sample guide merge diagnostics are written next to the merged H5AD files.

Pipeline Architecture

1. **Prepare samples:** Parse ENA metadata and group SRR runs by physical 10x well.
2. **Build references:** Create cDNA and KITE guide indices from Ensembl GRCh38 and `features.tsv`.
3. **Count reads:** Run `kb count` separately for mRNA and sgRNA FASTQs for each physical sample.
4. **Merge modalities:** Align guide counts to filtered mRNA cell barcodes.
5. **Concatenate samples:** Add sample-specific barcode suffixes and write one unified H5AD.

Key design point: sample-aware grouping prevents barcode collisions between separate physical wells.

Development Issues Solved

- ▶ **ENA FASTQs missing technical reads:** Historical ENA FASTQs for this dataset do not include the barcode/UMI reads. Current workaround: SRA download with technical reads included. We notified ENA; they are aware and plan to fix it, so future runs may use ENA directly.
- ▶ **Sample grouping:** SRRs and lanes must be grouped by physical well before counting. This is currently parsed from ENA library names; later versions may autodetect this.
- ▶ **Barcode collisions:** Final concatenation adds sample suffixes so identical 10x barcodes from different wells remain distinct.
- ▶ **GEX/KITE matching:** Guide barcodes are not exactly the same as expression barcodes in this dataset. The pipeline applies the required KITE barcode correction before alignment.

Pipeline Execution

```
1 nextflow run main.nf \  
2   -profile slurm,singularity \  
3   --fastq_dir $HPS_PATH/perturb_seq_fastq/SAMN40972597 \  
4   --metadata_tsv $HPS_PATH/perturb_seq_fastq/SAMN40972597/ena_metadata.tsv \  
5   --outdir $HPS_PATH/perturb_seq_fastq/results \  
6   --chemistry 10xv3 \  
7   --transcriptome_fa $HPS_PATH/cache/reference/Homo_sapiens.GRCh38.dna.  
    primary_assembly.fa.gz \  
8   --gtf $HPS_PATH/cache/reference/Homo_sapiens.GRCh38.111.gtf.gz \  
9   --features_tsv $HPS_PATH/perturb_seq_fastq/SAMN40972597/features.tsv
```

- ▶ The sample sheet is generated automatically from ENA metadata.
- ▶ mRNA and sgRNA workflows run in parallel per sample.
- ▶ Guide counts are merged from unfiltered KITE output, then aligned to filtered mRNA cells.

First Dataset: Nadig 2025 Jurkat

- ▶ Processed from raw FASTQs into a unified H5AD.
- ▶ The comparison was run against the author-provided H5AD.
- ▶ Analysis compares shared cells and shared genes after barcode and gene-name alignment.
- ▶ The main question: does the reprocessed dataset preserve expression signal, and how many guide calls match the original labels?

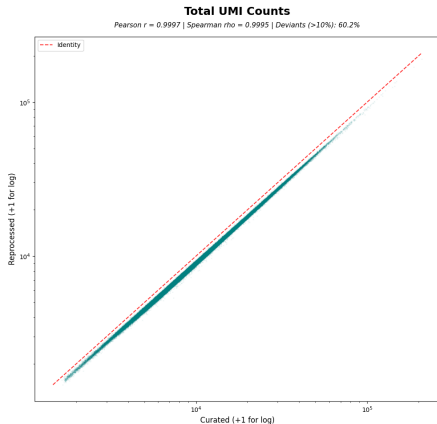
Short answer: expression alignment is excellent; guide recovery is currently limited by strict dual-probe calling.

Expression Agreement: Summary

Metric	Pearson	Spearman	Deviant cells/genes
Total UMI counts per cell	0.9997	0.9995	60.18%
Detected genes per cell	0.9999	0.9998	0.01%
Mean expression per gene	0.9588	0.9523	13.39%
Gene dropout rate	0.9598	0.9523	7.48%

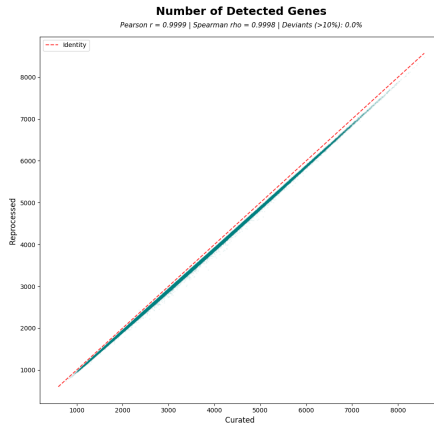
- ▶ Cell-level expression structure is almost identical to the author-processed dataset.
- ▶ Median cell-wise expression correlation is 0.9597.
- ▶ The main total-count difference is scale, not loss of biological structure.

Cell-Level Expression Agreement



Correlation of total UMI counts per cell. Kallisto+BusTools (Reprocessed) generally identifies more UMIs per cell, especially in the high-sensitivity regime, likely due to transcript-level mapping vs gene-level quantification.

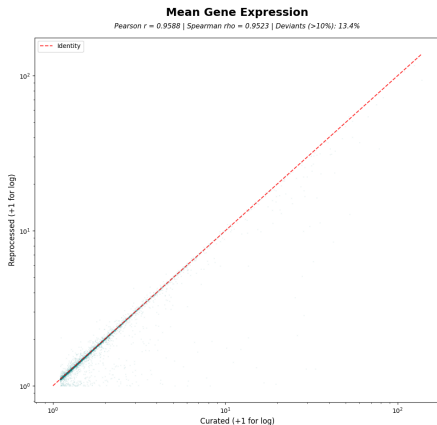
Total UMI counts



Correlation of unique genes detected per cell. The reprocessed pipeline recovers significantly more unique transcripts in a subset of cells, indicating higher detection sensitivity.

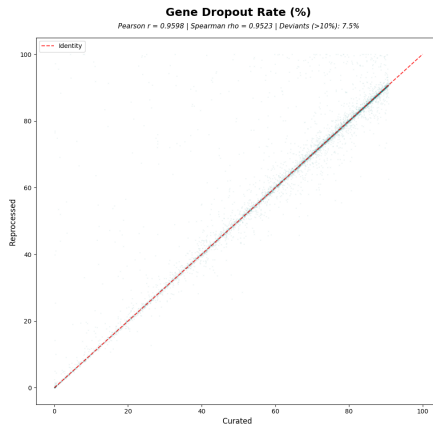
Number of detected genes

Gene-Level Expression Agreement



Average expression level for each shared gene. High correlation confirms that the biological signal is preserved across quantification methods.

Mean gene expression



Percentage of cells where a gene has zero counts. Lower values in Reprocessed indicate higher sensitivity for those genes.

Gene dropout rate

Guide Recovery: Current Result

- ▶ The comparison uses a strict guide call: two positive guides, both targeting the same gene, with at least 5 counts each.
- ▶ Under this rule, the reprocessed data recovers 164,757 of 214,522 curated dual-guide cells.
- ▶ Exact guide-pair agreement: **76.80%**.
- ▶ Target-level agreement: **76.80%**.
- ▶ The missing calls are mostly cells where only one of the two curated probes passes the 5-count threshold.

Interpretation: current guide recovery is about threshold/calling behaviour, not a failure of the expression pipeline.

Guide Signal Diagnostics

Threshold	Both curated probes	Exactly one probe	Neither probe
≥ 1 count	173,043	41,332	147
≥ 2 counts	170,245	44,038	239
≥ 5 counts	164,752	46,288	3,482
≥ 10 counts	158,854	47,254	8,414

- ▶ Across all reprocessed cells, 539,208 have at least one guide with 5+ counts.
- ▶ 487,177 have two or more guides with 5+ counts.
- ▶ 302,165 receive a dual same-target guide call before restricting to curated shared cells.

What We Have Now

- ▶ A working, reproducible FASTQ-to-H5AD pipeline for perturb-seq datasets with GEX and guide modalities.
- ▶ A comparison workflow that checks cell-level expression, gene-level expression, dropout, and perturbation labels.
- ▶ First results showing very strong expression agreement with the author-processed data.
- ▶ A clear guide-calling question to tune next: how strict should dual-probe recovery be?

Next Steps

- ▶ **Guide recovery:** Experiment with guide recovery settings. The current 5+5 threshold may be too strict.
- ▶ **Scale-out:** Run the pipeline on all available perturb-seq datasets.
- ▶ **Downstream analysis:** Run distance metrics between genes, plus the existing DEA and GSEA workflows.

Questions

Thank you